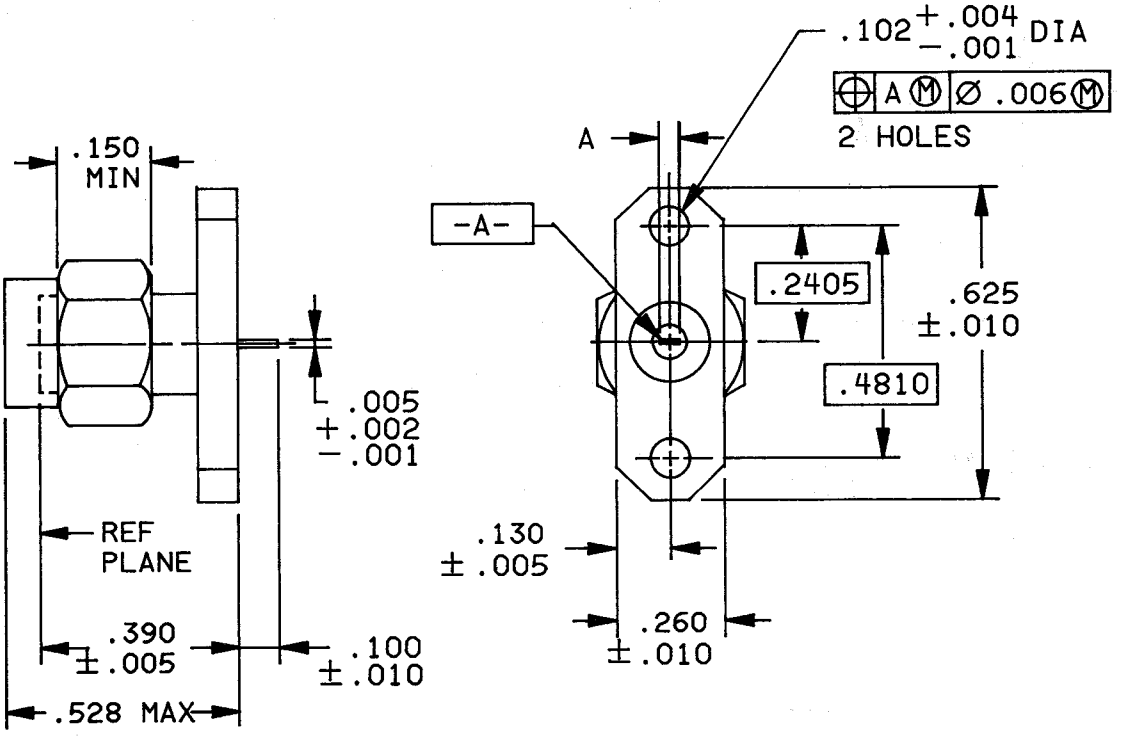
MILITARY SPECIFICATION SHEET

CONNECTOR, COAXIAL, RADIO FREQUENCY, STRIP OR MICROSTRIP TRANSMISSION LINE, SERIES SMA (PIN CONTACT, FLANGE MOUNTED TAB TERMINAL RECEPTACLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83517.



PART NO.	"A" DIM
-31001	.020 ±.002
-31002	.050 ±.001

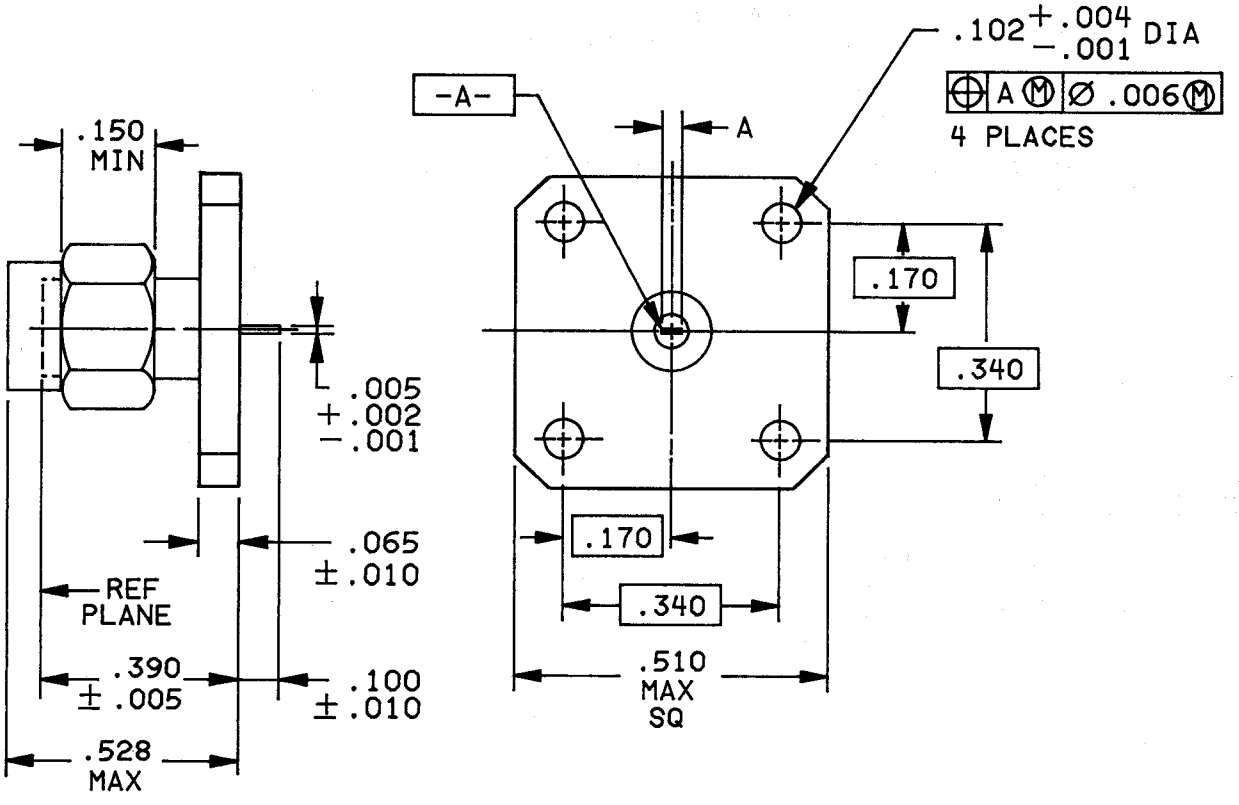
INCHES	MM	INCHES	MM
.001	.03	.100	2.54
.002	.05	.102	2.59
.004	.10	.130	3.30
.005	.13	.150	3.81
.006	.15	.2405	6.11
.010	.25	.260	6.60
.020	.51	.390	9.91
.050	1.27	.4810	12.22
.065	1.65	.528	13.41
		.625	15.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 1. Series SMA, pin contact, (2 hole) flange mounted receptacle.

FSC 5935

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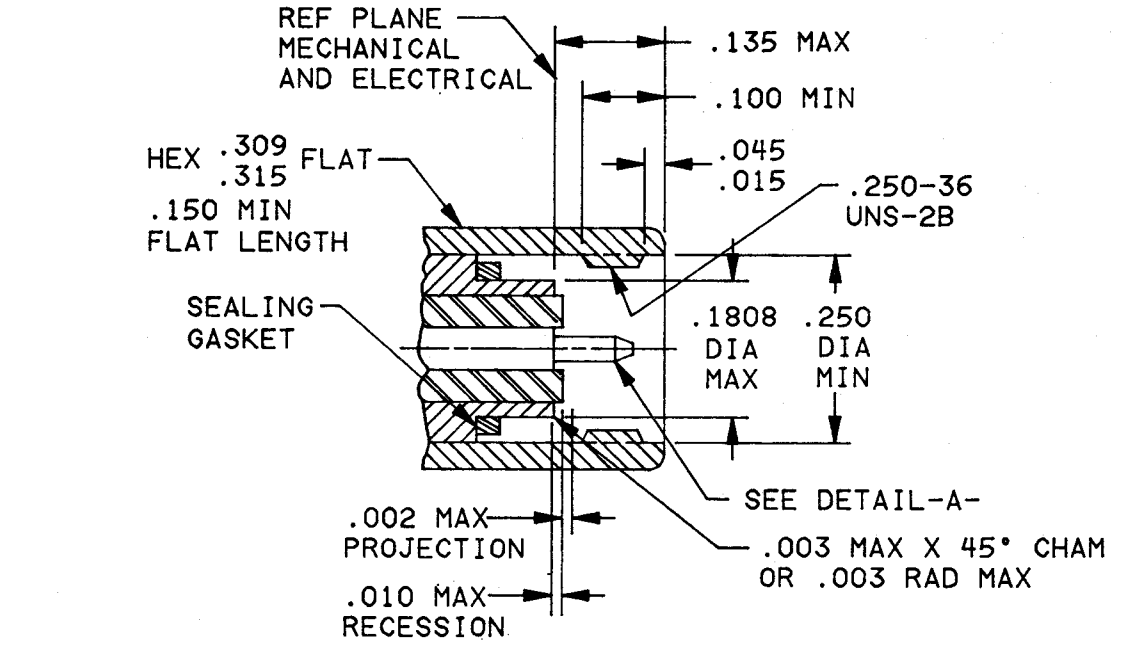
PART NO.	"A" DIM
-31003	.020 ±.002
-31004	.050 ±.001

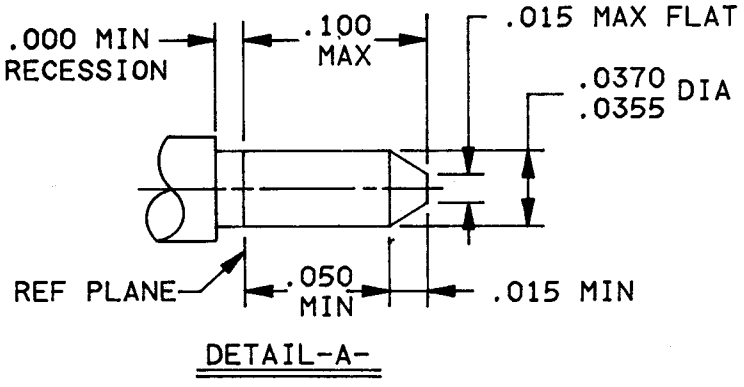
INCHES	MM	INCHES	MM
.001	.03	.065	1.65
.002	.05	.100	2.54
.004	.10	.102	2.59
.005	.13	.170	4.32
.006	.15	.340	8.64
.010	.25	.390	9.91
.020	.51	.510	12.95
.050	1.27	.528	13.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 2. Series SMA, pin contact, (4 hole) flange mounted receptacle.

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INCHES	MM
.002	.05
.003	.08
.015	.38
.0355	.901
.0370	.940
.045	1.14
.050	1.27
.100	2.54
.135	3.43
.150	3.81
.1808	4.592
.250	6.35
.309	7.85
.315	8.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 3. Mating dimensions for pin terminations.

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ENGINEERING PARAMETERS:

Nominal impedance: 50 ohms.

Voltage rating: 600 Vrms maximum at sea level.

150 Vrms maximum at 70,000 feet.

Frequency range: 0 to 18.0 GHz.

Temperature rating: -65° to 105°C.

REQUIREMENTS:

Design and construction: See figures 1 through 3 and table I.

Force to engage and disengage:

Torque - 2 inch-pounds maximum.

Longitudinal force - Not applicable.

Coupling proof torque: 15 inch-pounds minimum.

Inspection note: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

TABLE I. Part number and characteristics.

Part no. M83517/02-	Connector figure no.	Characteristics
31001	1	2 hole flange mount. .020 wide tab.
31002	1	2 hole flange mount. .050 wide tab.
31003	2	4 hole flange mount. .020 wide tab.
31004	2	4 hole flange mount. .050 wide tab.
32001	1	2 hole flange mount. .020 wide tab.
32002	1	2 hole flange mount. .050 wide tab.
32003	2	4 hole flange mount. .020 wide tab.
32004	2	4 hole flange mount. .050 wide tab.

Permeability of nonmagnetic materials: Applicable.

Seal:

Hermetic sealed connectors: Not applicable.

Pressurized and weatherproof connectors: Not applicable.

Insulation resistance: 5,000 megohms minimum.

Center contact retention:

Axial force: 6 pounds minimum.

Torque: 4 inch-ounces minimum.

Dielectric withstanding voltage: Applicable, test condition I.

Test voltage 1,000 Vrms.

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Corrosion: Applicable, test condition B.

Voltage standing wave ratio (VSWR):

Test frequency range: From .5 to 18.0 GHz.

Swept frequency VSWR test setup:

Step 1: See basic specification.

Step 2: VSWR shall be less than 1.080 +.005 frequency (frequency in GHz).

Item 11p VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Item 11j VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Step 3: VSWR shall be less than (to be determined).

Test fixture - See figure 1 of basic specification.

Step 4: VSWR test shall be less than (to be determined).

RF transmission loss: (To be determined).

RF leakage: Not applicable.

Connector durability:

Interface:

500 cycles minimum at 12 cycles/minute maximum rate.

Connector shall meet contact gaging and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable

Thermal shock: Applicable, test condition A.

Moisture resistance: Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

RF high potential withstanding voltage:

At a frequency between 5 to 7.5 MHz.

Leakage current - Not applicable.

RF voltage - 1,000 Vrms.

Coupling mechanism retention force: 60 pounds minimum.

Part number: M83517/2-(dash number from table I.)

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Group qualification: See table II.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors 1/2/	Qualifies the following connectors
I	3†001	3†001 3†002 3†003 3†004

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same material only.

† Denotes finish.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, AT
Navy - SH, AS
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 85

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-83517/2 - Connector, Coaxial, Radio Frequency, Strip or Microstrip Transmission Line, Series SMA

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

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